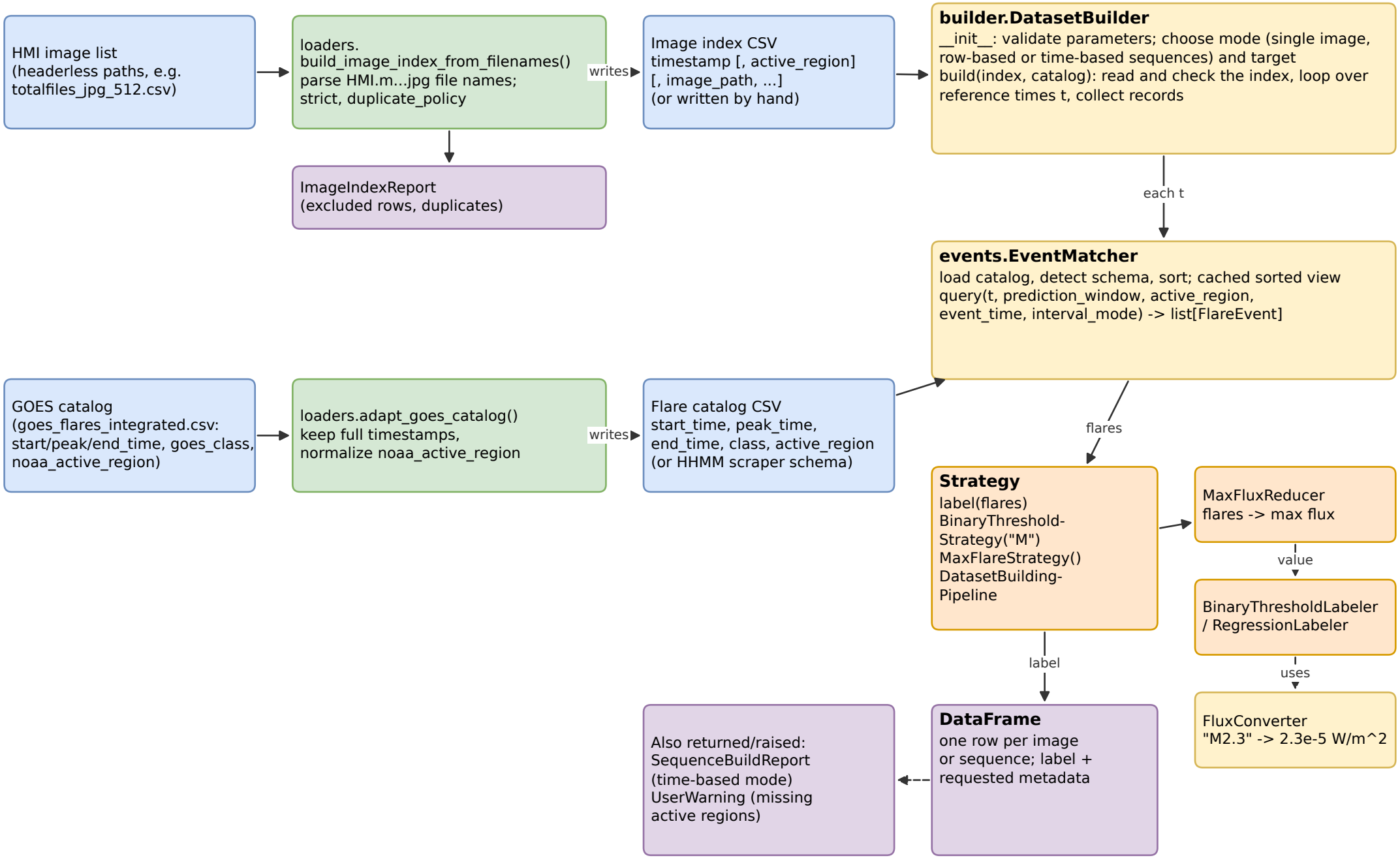
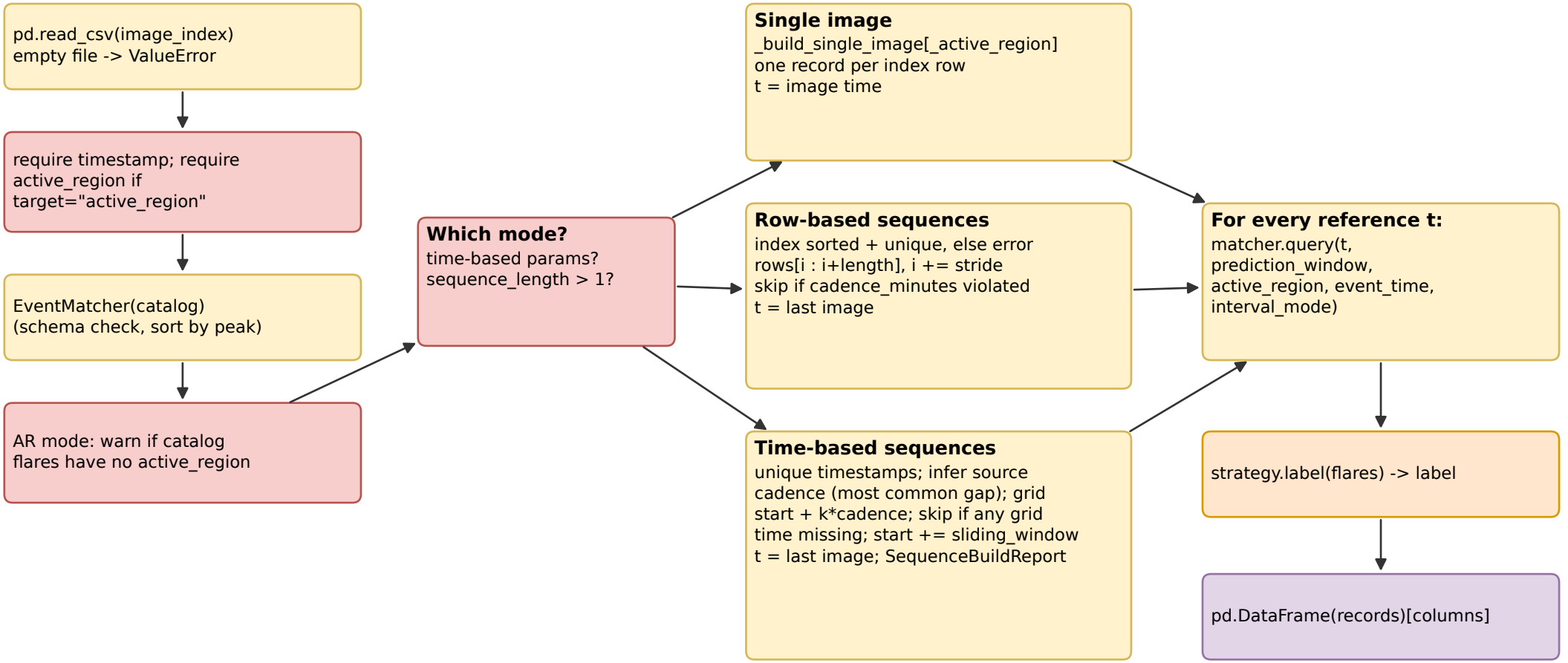
flare-indexer: components and data flow

Real inputs -> loaders -> CSV inputs -> DatasetBuilder -> EventMatcher -> strategy -> labeled table



DatasetBuilder.build(): control flow

Mode is fixed in __init__; build() dispatches to one of six private builders (three modes x two targets)

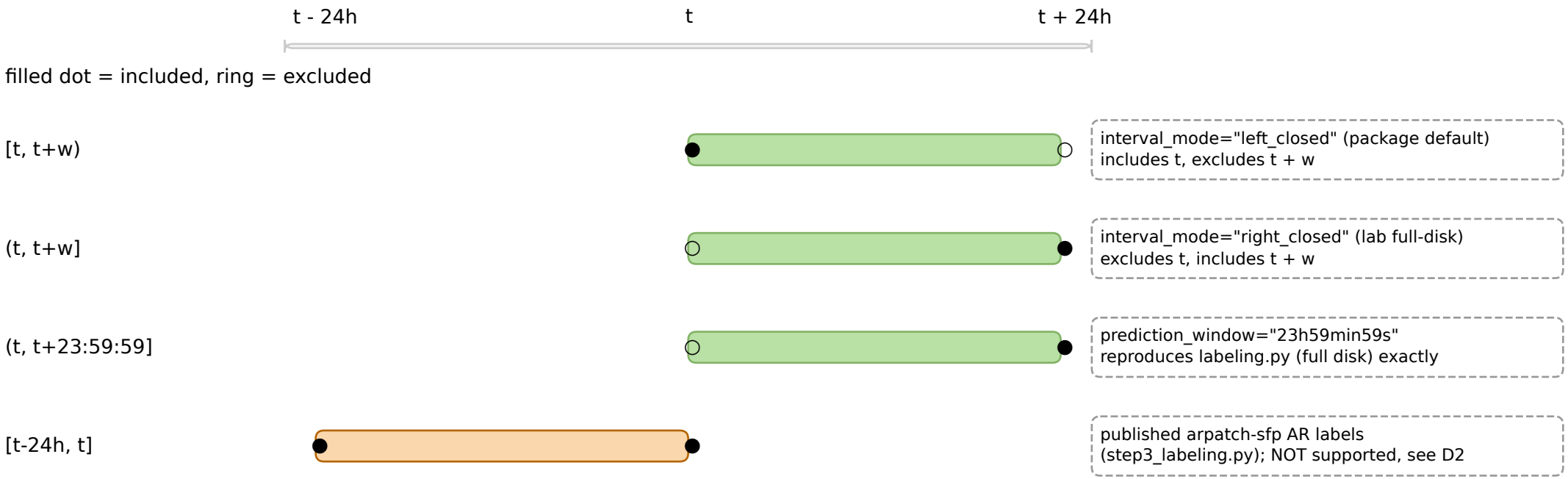


target="active_region" variants of every mode

normalize region (13600 / 13600.0 / "13600" -> "13600"); query only that region's flares
image region missing -> missing_active_region: "negative" (+UserWarning) / "drop" / "error"
sequence with two different regions -> ValueError; any missing region in a sequence -> treated as missing

Prediction window semantics

Which flares count for an image at time t (w = prediction_window). Times are compared as given (no TAI/UTC conversion).



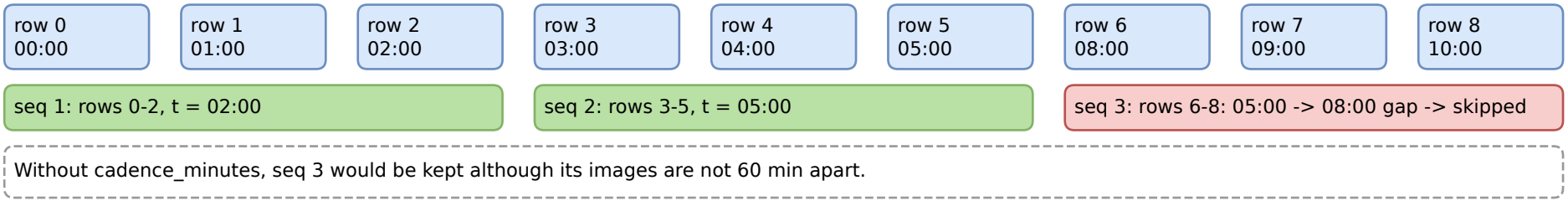
event_time selects the flare time that must fall inside the window:
  "peak" (default) -> peak_time
  "start" -> start_time (lab convention)
Sequences: t = sequence_end (last image of the sequence).

Example from the real full-disk data (2010-2018, 63,649 images):
image 2014-08-23 12:00, M5.9 flare starts 2014-08-24 12:00 (= t + 24h)
  (t, t+24h]   -> label 1
  (t, t+23:59:59] -> label 0 (legacy)
Six images differ this way; with 23h59min59s all 63,649 agree.

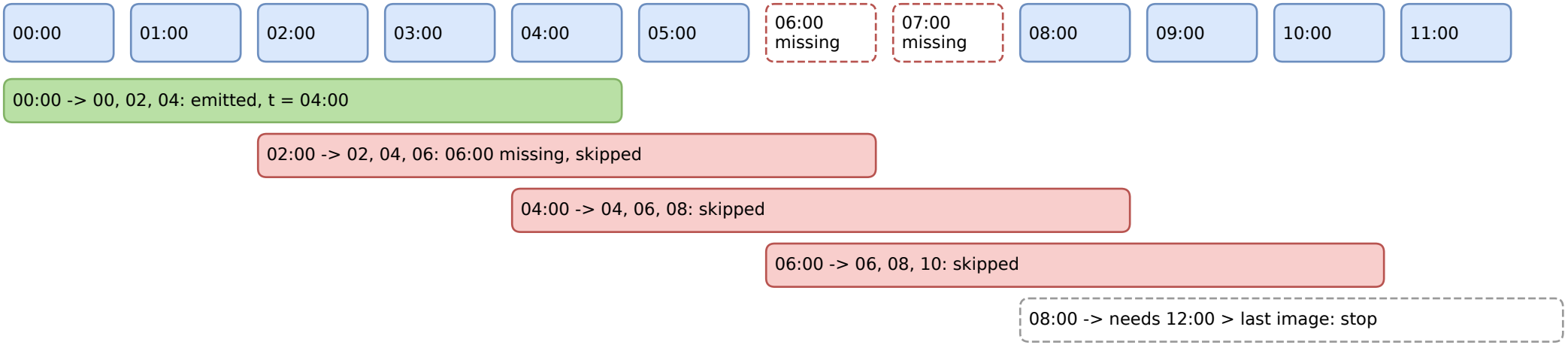
Sequence construction

Row-based uses positions in the index; time-based uses an exact timestamp grid and skips incomplete grids

Row-based: sequence_length=3, stride=3, cadence_minutes=60 (index must be sorted and unique)



Time-based: cadence="2h", observation_window="6h" (3 images), sliding_window="2h"; inferred source cadence = 1h



Rules (builder._time_based_candidates)

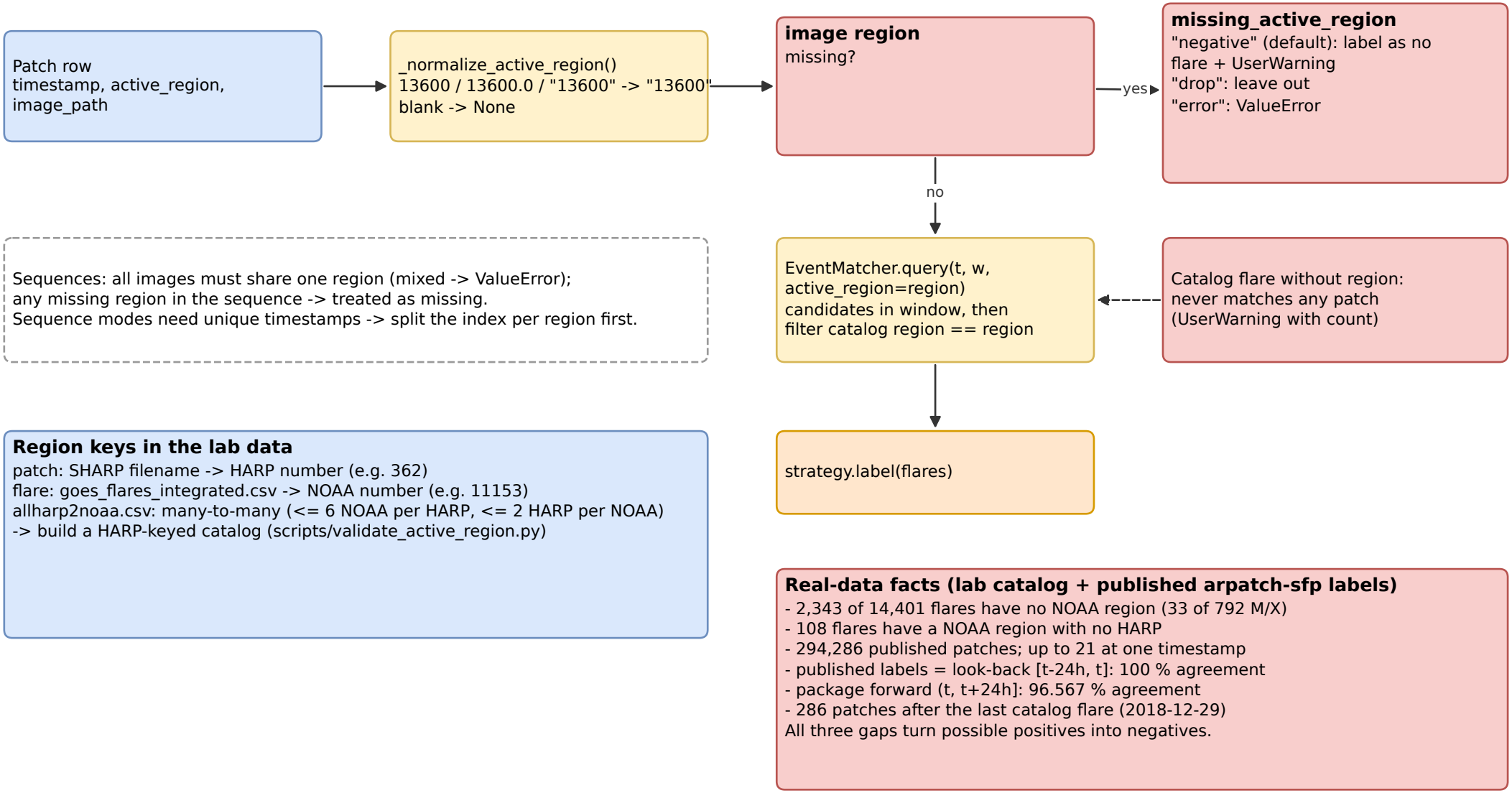
cadence >= inferred cadence and an exact multiple of it
observation_window, sliding_window: multiples of cadence
start (optional) must be an existing image time
candidates while start + observation_window - cadence <= last image
missing grid time -> skip; never interpolate or substitute
return_report=True -> SequenceBuildReport (counts, first 50 skips)

Output row per emitted sequence

sequence_start, sequence_end, timestamps (list), n_images, [active_region], label, + every other index column as a list

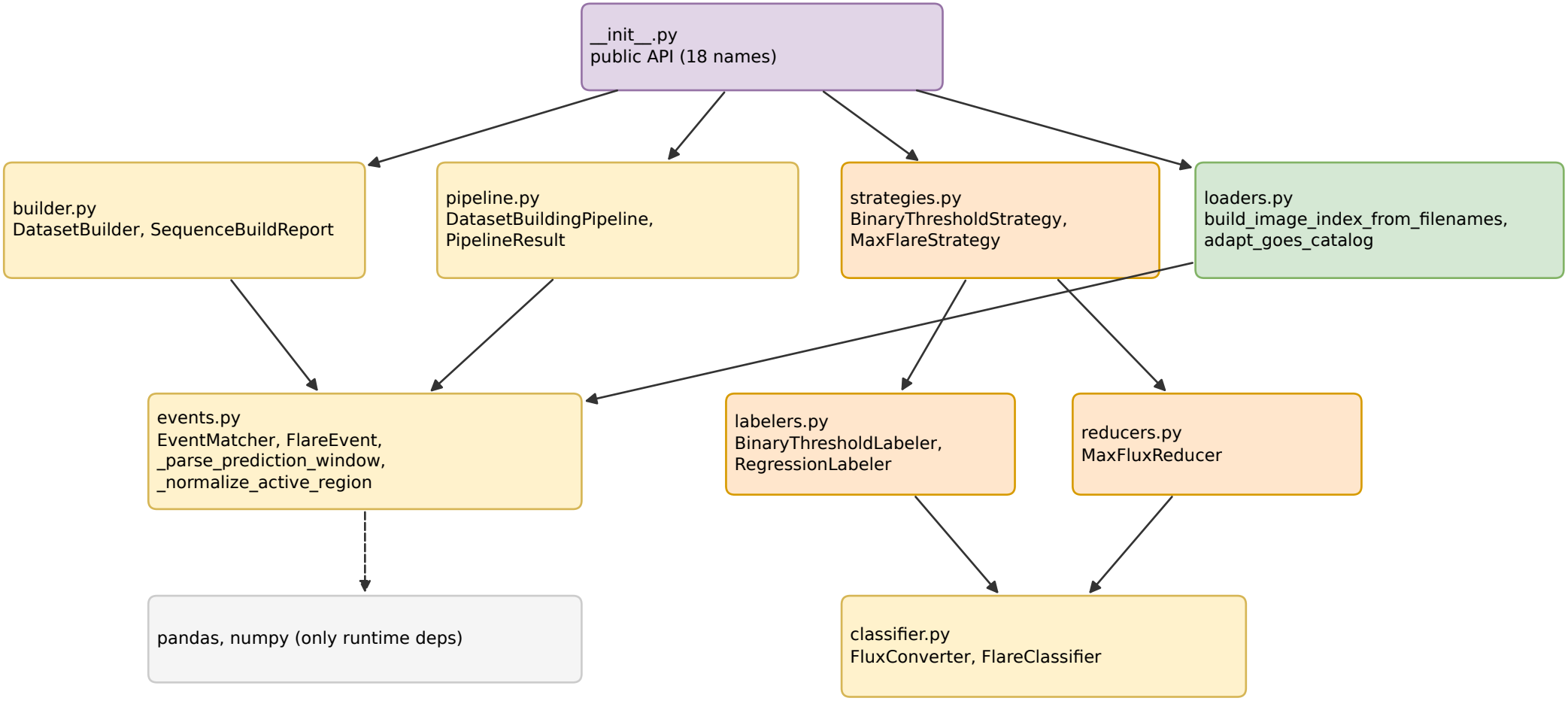
Active-region matching and missing data

target="active_region": a flare counts only if its region key equals the patch's region key



Package modules, dependencies and verification

src/flare_indexer/ (public API re-exported from __init__.py); arrows point from user to dependency*



Scripts, examples, committed results

scripts/validate_full_disk.py -> real full-disk data vs legacy labels
scripts/validate_active_region.py -> published AR labels (HARP key)
examples/01..06_*.py + examples/data (synthetic)
docs/validation/*.json, *.txt (committed results)

tests/ (pytest, CI on Python 3.10-3.13)

unit: events, builder, loaders, strategies, pipeline, ...
test_windows_and_missing_regions.py: duration windows, AR policy
test_examples.py: every example runs; sample data regenerates